Multioperator splitting methods for a class of nonmonotone inclusion problems

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Outline

General overview

Problem setting

$$\text{Find } x \in \mathcal{H} \text{ such that } 0 \in A_1(x) + A_2(x) + \cdots + A_n(x)$$

- $\mathcal{H}$ is a real Hilbert space
- $A_i : \mathcal{H} \rightrightarrows \mathcal{H}$ is a set valued operator, $n \geq 3$

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Main research question

How far can monotone splitting theory be extended **beyond monotonicity**?

What is known?

Classical setting:

$$A_i \text{ is maximally monotone} \qquad (i = 1, \dots, n).$$

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$$A_i \text{ is maximally monotone} \qquad (i = 1, \ldots, n).$$

Definition (Maximal monotone)

An operator $A : \mathcal{H} \rightrightarrows \mathcal{H}$ is monotone if

$$\langle x - y, u - v \rangle \geq 0 \qquad \forall (x, u), (y, v) \in \operatorname{gra} A.$$

It is maximally monotone if its graph has no proper monotone extension.

Example

If $f : \mathcal{H} \to (-\infty, +\infty]$ is proper, closed, and convex, then

$$\partial f \text{ is maximally monotone.}$$

Splitting algorithms

Leverage desirable properties of the individual terms:

$$A_1, \ldots, A_n.$$

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Backward evaluation

$$J_{\gamma A_i} := (\mathrm{Id} + \gamma A_i)^{-1}$$

Forward evaluation

$$x \mapsto A_i x \qquad \text{when } A_i \text{ is single-valued.}$$

Two-operator splitting (1)

For

$$0 \in A_1 x + A_2 x,$$

Douglas–Rachford splitting:

$$\begin{cases} x_1^k = J_{\gamma A_1}(z^k), \\ x_2^k = J_{\gamma A_2}(2x_1^k - z^k), \\ z^{k+1} = z^k + \lambda_k (x_2^k - x_1^k). \end{cases}$$

Main idea

Use $J_{\gamma A_1}$ and $J_{\gamma A_2}$, not the resolvent of $A_1 + A_2$.

Two-operator splitting (2)

For

$$0 \in Ax + Bx,$$

with A backward and B forward:

$$\begin{cases} x^k = J_{\gamma A}\big(z^k - \gamma B z^k\big), \\ z^{k+1} = z^k + \lambda_k\big(x^k - z^k\big). \end{cases}$$

Forward–Backward

Backward step for A, forward step for B.

Recent works on multioperator splitting

1. product space Douglas–Rachford methods (Campoy, 2022)
2. graph and distributed extensions (Bredies–Chenchene–Naldi, 2024)
3. graph-devised forward–backward methods*
   (Aragón-Artacho–Campoy–López-Pastor, 2025)
4. full characterization of forward–backward splittings*
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Common setting

The convergence theory relies on monotonicity.

Back to the main question

Problem setting

$$\text{Find } x \in \mathcal{H} \text{ such that } 0 \in A_1(x) + A_2(x) + \cdots + A_n(x)$$

Main research question

Can splitting methods still converge when the operators are **not necessarily monotone**?

Outline

Product-space formulation

Following Campoy/Kruger, set

$$\mathbf{x} = (x_1, \ldots, x_{n-1}) \in \mathcal{H}^{n-1}, \qquad \mathbf{D}_{n-1} := \{(x, \ldots, x) : x \in \mathcal{H}\}.$$

Define

$$\begin{aligned} \mathbf{F}(\mathbf{x}) &:= A_1(x_1) \times \cdots \times A_{n-1}(x_{n-1}) \\ \mathbf{G}(\mathbf{x}) &:= \mathbf{K}(\mathbf{x}) + N_{\mathbf{D}_{n-1}}(\mathbf{x}), \end{aligned}$$

where

$$\mathbf{K}(\mathbf{x}) := \frac{1}{n-1} A_n(x_1) \times \cdots \times \frac{1}{n-1} A_n(x_{n-1}).$$

Reduction

$$0 \in A_1 x + \cdots + A_n x \quad \Longleftrightarrow \quad 0 \in \mathbf{F}(\mathbf{x}) + \mathbf{G}(\mathbf{x}).$$

Resolvents of **F** and **G**

Let

$$\bar{x} := \frac{1}{n-1}\sum_{i=1}^{n-1} x_i, \qquad \boldsymbol{\Delta}_{n-1}(x) := (x, \dots, x).$$

Then

$$\begin{aligned} J_{\gamma\mathbf{F}}(\mathbf{x}) &= \left(J_{\gamma A_1}(x_1), \dots, J_{\gamma A_{n-1}}(x_{n-1})\right), \\ J_{\gamma\mathbf{G}}(\mathbf{x}) &= \boldsymbol{\Delta}_{n-1}\left(J_{\frac{\gamma}{n-1}A_n}(\bar{x})\right). \end{aligned}$$

Resolvents of **F** and **G** reduce to resolvents of the A_i's.

Multioperator Douglas–Rachford

Apply Douglas–Rachford to

$$0 \in \mathbf{F}(\mathbf{x}) + \mathbf{G}(\mathbf{x}).$$

$$\begin{cases} \mathbf{y}^k \in J_{\gamma\mathbf{F}}(\mathbf{x}^k), \\ \mathbf{z}^k \in J_{\gamma\mathbf{G}}(2\mathbf{y}^k - \mathbf{x}^k), \\ \mathbf{x}^{k+1} = \mathbf{x}^k + \kappa(\mathbf{z}^k - \mathbf{y}^k). \end{cases}$$

Unwrapped algorithm

$$
\begin{cases}
y_i^k \in J_{\gamma A_i}(x_i^k), & i = 1, \dots, n-1, \\
z^k \in J_{\frac{\gamma}{n-1} A_n}\left(\dfrac{1}{n-1} \displaystyle\sum_{i=1}^{n-1} (2y_i^k - x_i^k) \right), & \\
x_i^{k+1} = x_i^k + \kappa (z^k - y_i^k), & i = 1, \dots, n-1.
\end{cases}
$$

Structure

Parallel evaluations of $A_1, \dots, A_{n-1}$, followed by one evaluation of A_n.

Two nonmonotone classes

Definition

Let $A : \mathcal{H} \rightrightarrows \mathcal{H}$ and $\sigma \in \mathbb{R}$.

A is σ-monotone if

$$\langle x - y, u - v \rangle \geq \sigma \|x - y\|^2 \qquad \forall (x, u), (y, v) \in \operatorname{gra} A.$$

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Note: Nonmonotone regime is when $\sigma < 0$.

Generalized monotone case

Two-operator theory for generalized monotone case (Dao–Phan, 2019; Giselsson–Moursi, 2020)

1. For
$$0 \in F + G,$$
where F is maximal σ_F-monotone, and G is maximal σ_G-monotone.
2. DR convergence is known when $\sigma_F + \sigma_G > 0$.

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2. DR convergence is known when $\sigma_F + \sigma_G > 0$.

Applied naively to $\mathbf{F} + \mathbf{G}$

Assume

$$A_i \text{ is maximal } \sigma_i\text{-monotone.}$$

One obtains a condition of the form

$$\underbrace{\min_{i=1,\ldots,n-1} \sigma_i}_{\sigma_{\mathbf{F}}} + \underbrace{\frac{\sigma_n}{n-1}}_{\sigma_{\mathbf{G}}} > 0.$$

Improved result

Theorem (informal)

With suitable stepsize, the multioperator Douglas–Rachford method converges under

$$\sum_{i=1}^{n} \sigma_i > 0.$$

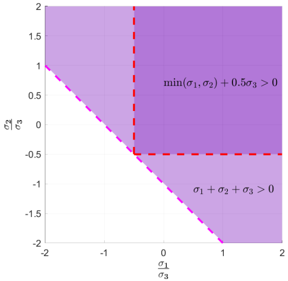

Figure: $\boxed{\min\{\sigma_1, \sigma_2\} + \frac{\sigma_3}{2} > 0}$ vs. $\boxed{\sigma_1 + \sigma_2 + \sigma_3 > 0}$ when $\sigma_3 > 0$.

Improved result

Comonotone case

Two-operator theory for comonotone case(Bartz–Dao–Phan, 2022)

If F is σ_F-comonotone and G is σ_G-comonotone, DR convergence is known when

$$\sigma_F + \sigma_G > 0.$$

If $\sigma_G < 0$, then equivalently,

$$\frac{1}{\sigma_F} + \frac{1}{\sigma_G} < 0$$

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Applied naively to $\mathbf{F} + \mathbf{G}$

If A_i is maximal σ_i-comonotone with $\sigma_n < 0$, the condition reduces to

$$\underbrace{\frac{1}{\min\limits_{i=1,\ldots,n-1} \sigma_i}}_{\frac{1}{\sigma_{\mathbf{F}}}} + \underbrace{\frac{1}{(n-1)\sigma_n}}_{\frac{1}{\sigma_{\mathbf{G}}}} < 0.$$

Improved result

Theorem (informal)

With suitable stepsize, the multioperator Douglas–Rachford method converges under

$$\sum_{i=1}^{n} \frac{1}{\sigma_i} < 0.$$

Takeaway

Douglas–Rachford survives beyond monotonicity.

Limitation

- product space geometry + "weighted Fejér analysis".
- The proof strongly uses the special structure

$$A_1, \ldots, A_{n-1} \quad \longrightarrow \quad A_n.$$

The previous analysis is powerful, but tailored to this DR structure.

Next question

Can we extend this to a more general splitting framework?

Outline

General framework

General Problem setting

$$\text{Find } \mathbf{x} \in \boldsymbol{\mathcal{H}} \text{ such that } 0 \in \mathbf{A}(\mathbf{x}) + N_{\boldsymbol{\Omega}}(\mathbf{x}). \tag{P}$$

- Here,

$$\boldsymbol{\Omega} \subseteq \boldsymbol{\mathcal{H}} \quad \text{is a closed linear subspace.}$$

- $\mathbf{A} : \boldsymbol{\mathcal{H}} \rightrightarrows \boldsymbol{\mathcal{H}}$ is a set-valued operator

General framework

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- $\mathbf{A} : \boldsymbol{\mathcal{H}} \rightrightarrows \boldsymbol{\mathcal{H}}$ is a set-valued operator
- **Special case:** Pierra's product space reformulation

$$\begin{aligned} \mathbf{A} &= \operatorname{diag}(A_1, \ldots, A_n), \\ \boldsymbol{\Omega} &= \{(x, \ldots, x) : x \in \mathcal{H}\} \\ \mathbf{H} &= \mathcal{H} \times \cdots \times \mathcal{H}. \end{aligned}$$

Two linear operators

- **Parametrization of Ω:** Choose a bounded linear operator $\mathbf{M} : \mathcal{H}' \to \mathcal{H}$ with closed range such that

$$\ker(\mathbf{M}^*) = \mathbf{\Omega}.$$

Then

$$N_{\mathbf{\Omega}}(\mathbf{x}) = \mathbf{\Omega}^{\perp} = \operatorname{ran}(\mathbf{M}).$$

Two linear operators

- **Parametrization of $\boldsymbol{\Omega}$:** Choose a bounded linear operator $\mathbf{M} : \boldsymbol{\mathcal{H}}' \to \boldsymbol{\mathcal{H}}$ with closed range such that

$$\ker(\mathbf{M}^*) = \boldsymbol{\Omega}.$$

Then

$$N_{\boldsymbol{\Omega}}(\mathbf{x}) = \boldsymbol{\Omega}^{\perp} = \operatorname{ran}(\mathbf{M}).$$

- **"Preconditioner":** Choose a bounded linear operator $\mathbf{L} : \boldsymbol{\mathcal{H}} \to \boldsymbol{\mathcal{H}}$ such that

$$\mathbf{L}(\boldsymbol{\Omega}) \subseteq \boldsymbol{\Omega}^{\perp}.$$

Reformulation

Theorem (reformulation)

Let $\gamma > 0$. Then $\mathbf{x}$ solves

$$0 \in \mathbf{A}(\mathbf{x}) + N_{\boldsymbol{\Omega}}(\mathbf{x})$$

if and only if there exists $\mathbf{z} \in \boldsymbol{\mathcal{H}}'$ such that

$$\begin{cases} \mathbf{M}\mathbf{z} \in \gamma \mathbf{A}(\mathbf{x}) + \mathbf{L}\mathbf{x}, \\ \mathbf{M}^*\mathbf{x} = 0 \end{cases}$$

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Proof idea

Use

$$N_{\mathbf{\Omega}}(\mathbf{x}) = \mathbf{\Omega}^{\perp} = \operatorname{ran}(\mathbf{M}), \qquad \mathbf{L}\mathbf{x} \in \mathbf{\Omega}^{\perp} \quad (\mathbf{x} \in \mathbf{\Omega}).$$

Proposed algorithm

Input: Choose $\gamma > 0$, relaxation parameters $\{\lambda_k\}_{k\geq 0} \subset (0,\infty)$, and $\mathbf{z}^0 \in \boldsymbol{\mathcal{H}}'$.

For $k = 0, 1, 2, \ldots$, compute:

$$\boxed{\begin{array}{ll} \text{Step 1:} & \text{find } \mathbf{x}^k \in \boldsymbol{\mathcal{H}} \quad \text{s.t.} \quad \mathbf{M}\mathbf{z}^k \in \gamma \mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k, \\[1ex] \text{Step 2:} & \mathbf{z}^{k+1} = \mathbf{z}^k - \lambda_k \mathbf{M}^* \mathbf{x}^k. \end{array}}$$

Note: The $\mathbf{x}^k$-step is implicit

Example 1: Finding zero of $A_1 + A_2$

Take $\mathbf{\Omega} = \{(x, x) : x \in \mathcal{H}\}$.

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} A_1 x_1 \\ A_2 x_2 \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} \\ -\mathrm{Id}_{\mathcal{H}} \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} & 0 \\ -2\,\mathrm{Id}_{\mathcal{H}} & \mathrm{Id}_{\mathcal{H}} \end{bmatrix}.$$

Then $\ker(\mathbf{M}^*) = \mathbf{\Omega}$ and

$$\mathbf{M}\mathbf{z}^k \in \gamma \mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k$$

becomes

$$\begin{bmatrix} z^k \\ -z^k \end{bmatrix} \in \gamma \begin{bmatrix} A_1 x_1^k \\ A_2 x_2^k \end{bmatrix} + \begin{bmatrix} x_1^k \\ -2x_1^k + x_2^k \end{bmatrix}.$$

Hence

$$\begin{cases} z^k \in (\gamma A_1 + \mathrm{Id}) x_1^k, \\ 2x_1^k - z^k \in (\gamma A_2 + \mathrm{Id}) x_2^k. \end{cases}$$

Recovered method

Thus,

$$\begin{cases} x_1^k \in J_{\gamma A_1}(z^k), \\ x_2^k \in J_{\gamma A_2}(2x_1^k - z^k), \\ z^{k+1} = z^k - \lambda_k(x_1^k - x_2^k). \end{cases}$$

That is,

$$z^{k+1} = z^k + \lambda_k(x_2^k - x_1^k).$$

Recovered method

This realization gives the Douglas–Rachford splitting for

$$0 \in A_1 x + A_2 x.$$

Example 2: Campoy's multioperator DR

Choose $\mathbf{M} = M \otimes \mathrm{Id}_{\mathcal{H}}$, and $\mathbf{L} = L \otimes \mathrm{Id}_{\mathcal{H}}$, where

$$M = \frac{1}{n-1}\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -1 & -1 & \cdots & -1 \end{bmatrix}$$

and

$$L = \begin{bmatrix} \frac{1}{n-1} & 0 & \cdots & 0 & 0 \\ 0 & \frac{1}{n-1} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \frac{1}{n-1} & 0 \\ -\frac{2}{n-1} & -\frac{2}{n-1} & \cdots & -\frac{2}{n-1} & 1 \end{bmatrix}.$$

Recovered method

These choices recover Campoy's product-space Douglas–Rachford method.

General algorithm

Input: Choose $\gamma > 0$, relaxation parameters $\{\lambda_k\}_{k\geq 0} \subset (0, \infty)$, and $\mathbf{z}^0 \in \boldsymbol{\mathcal{H}}'$.
For $k = 0, 1, 2, \ldots$, compute:

$$\text{Step 1: find } \mathbf{x}^k \in \boldsymbol{\mathcal{H}} \quad \text{s.t.} \quad \mathbf{M}\mathbf{z}^k \in \gamma \mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k,$$

$$\text{Step 2: } \mathbf{z}^{k+1} = \mathbf{z}^k - \lambda_k \mathbf{M}^* \mathbf{x}^k.$$

Forward interpretation of the algorithm

General algorithm

Input: Choose $\gamma > 0$, relaxation parameters $\{\lambda_k\}_{k\geq 0} \subset (0, \infty)$, and $\mathbf{z}^0 \in \mathcal{H}'$.
For $k = 0, 1, 2, \ldots$, compute:

$$\text{Step 1: find } \mathbf{x}^k \in \mathcal{H} \quad \text{s.t.} \quad \mathbf{M}\mathbf{z}^k \in \gamma\mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k,$$

$$\text{Step 2: } \mathbf{z}^{k+1} = \mathbf{z}^k - \lambda_k \mathbf{M}^*\mathbf{x}^k.$$

Define $\Phi_\gamma(\mathbf{w}) := (\gamma\mathbf{A} + \mathbf{L})^{-1}(\mathbf{w})$.

Algorithm in one line

If Φ_γ is single-valued, the algorithm can be written as

$$\boxed{\mathbf{z}^{k+1} = \mathbf{z}^k - \lambda_k \Psi_\gamma(\mathbf{z}^k)}$$

where

$$\Psi_\gamma(\mathbf{z}) := \mathbf{M}^*\Phi_\gamma(\mathbf{M}\mathbf{z})$$

Well-known result

Lemma (see Bauschke & Combettes, 2017)

If F is β-cocoercive, $\operatorname{zer}(F) \neq \emptyset$ and $\lambda_k \in [0, 2\beta]$ satisfies $\sum_{k=0}^{\infty} \lambda_k(2\beta - \lambda_k) = +\infty$. Define

$$z^{k+1} = z^k - \lambda_k F(z^k).$$

1. $\{z^k\}$ is Fejér monotone with respect to $\operatorname{zer}(F)$, i.e.,

$$\left\| z^{k+1} - z \right\| \leq \left\| z^k - z \right\| \quad \forall z \in \operatorname{zer}(F) \text{ and } k \geq 0.$$

2. $\{F(z^k)\}$ converges strongly to zero.
3. $z^k \rightharpoonup z^*$ for some $z^* \in \operatorname{zer}(F)$.
4. If $\liminf_{k\to\infty} \lambda_k(2\beta - \lambda_k) > 0$, then $\left\| F(z^k) \right\| = o(1/\sqrt{k})$.

Assumptions (1)

Define $\Phi_\gamma : \mathbf{\Omega}^\perp \rightrightarrows \boldsymbol{\mathcal{H}}$ by

$$\Phi_\gamma(\mathbf{w}) := (\gamma \mathbf{A} + \mathbf{L})^{-1}(\mathbf{w}).$$

[**Note:** $\mathbf{x}^k \in \Phi_\gamma(\mathbf{M}\mathbf{z}^k)$]

Assumption A.

The set

$$\Lambda := \left\{ \gamma > 0 : \Phi_\gamma \text{ is single-valued at every point in } \mathbf{\Omega}^\perp \right\}$$

is nonempty.

Assumptions (2)

Assumption B.

1. $\mathbf{A} : \mathcal{H} \rightrightarrows \mathcal{H}$ is $(\mathbf{U},\mathbf{V})$-semimonotone:

$$\langle \mathbf{x}-\mathbf{x}', \mathbf{y}-\mathbf{y}' \rangle \ \geq \ \langle \mathbf{x}-\mathbf{x}', \mathbf{U}(\mathbf{x}-\mathbf{x}') \rangle + \langle \mathbf{y}-\mathbf{y}', \mathbf{V}(\mathbf{y}-\mathbf{y}') \rangle$$

   for all $(\mathbf{x},\mathbf{y}), (\mathbf{x}',\mathbf{y}') \in \operatorname{gra}(\mathbf{A})$.

2. For $\mathbf{U}$, we assume that either
   - $\mathbf{U} = 0$; or
   - $\mathbf{U}_{11}$ is strongly monotone.

3. For $\mathbf{V}$, we assume that either
   - $\mathbf{V} = 0$; or
   - $\mathbf{V} \neq 0$ but $\mathbf{V}_{22} \succeq 0$, $\operatorname{ran}(\mathbf{V}_{22})$ is closed and $\operatorname{ran}(\mathbf{V}_{21}) \subseteq \operatorname{ran}(\mathbf{V}_{22})$.

[1] The block decomposition of $\mathbf{U}$ is with respect to $\mathbf{H} = \boldsymbol{\Omega} \oplus \boldsymbol{\Omega}^{\perp}$, i.e.,

$$\mathbf{U} = \begin{array}{c c} & \begin{array}{cc} \boldsymbol{\Omega} & \boldsymbol{\Omega}^{\perp} \end{array} \\ \begin{array}{c} \boldsymbol{\Omega} \\ \boldsymbol{\Omega}^{\perp} \end{array} & \begin{bmatrix} \mathbf{U}_{11} & \mathbf{U}_{12} \\ \mathbf{U}_{21} & \mathbf{U}_{22} \end{bmatrix} \end{array}.$$

Assumptions (3)

Define $\widehat{\boldsymbol{\Xi}}_\gamma := \mathbf{L}_s + \frac{1}{\gamma}\mathbf{L}^*(\mathbf{V}/\mathbf{V}_{22})\mathbf{L} + \gamma\mathbf{U}/\mathbf{U}_{11}$, and let

$$\Gamma := \left\{\gamma > 0 : \lambda_{\max}\left((\mathbf{L}_s^{1/2})^\dagger \left(\mathbf{L}_s - \widehat{\boldsymbol{\Xi}}_\gamma\right)(\mathbf{L}_s^{1/2})^\dagger\right) < 1\right\}$$

Assumption C.

1. $\Gamma \neq \emptyset$.
2. $\mathbf{L}_s := \frac{\mathbf{L}+\mathbf{L}^*}{2}$ satisfies
   - $\ker(\mathbf{L}_s) = \boldsymbol{\Omega}$.
   - $\mathbf{L}_s$ is strongly monotone on $\boldsymbol{\Omega}^\perp$.

Lemma

$$\gamma \in \Gamma \quad \Longrightarrow \quad \widehat{\boldsymbol{\Xi}}_\gamma \succeq \underbrace{\left\|\widehat{\boldsymbol{\Xi}}_\gamma^{1/2}\mathbf{M}\right\|^{-1/2}}_{\beta_\gamma}\mathbf{M}\mathbf{M}^*$$

Convergence Theorem

Algorithm:

$\boxed{\mathbf{z}^{k+1} = \mathbf{z}^k - \lambda_k \Psi_\gamma(\mathbf{z}^k)}$, where $\Psi_\gamma(\mathbf{z}) := \mathbf{M}^* \Phi_\gamma(\mathbf{M}\mathbf{z})$.

Theorem (Part 1)

Choose $\gamma \in \Lambda \cap \Gamma$ and let $\beta_\gamma := \left\| \widehat{\boldsymbol{\Xi}}_\gamma^{1/2} \mathbf{M} \right\|^{-1/2}$. Assume $\lambda_k \in [0, 2\beta_\gamma]$ satisfies $\sum_{k=0}^\infty \lambda_k(2\beta_\gamma - \lambda_k) = +\infty$. The following hold:

1. $\{\mathbf{z}^k\}$ is Fejér monotone with respect to $\operatorname{zer}(\Psi_\gamma)$.
2. $\Psi_\gamma(\mathbf{z}^k) = \mathbf{M}^* \mathbf{x}^k \to 0$.
3. $\mathbf{z}^k \rightharpoonup \mathbf{z}^*$ for some $\mathbf{z}^* \in \operatorname{zer}(\Psi_\gamma)$.
4. If $\liminf_{k\to\infty} \lambda_k(2\beta_\gamma - \lambda_k) > 0$, then $\left\|\Psi_\gamma(\mathbf{z}^k)\right\| = o(1/\sqrt{k})$.

Proof outline

It is enough to show that Ψ_γ is β_γ-cocoercive! That is,

$$\langle \Psi_\gamma(\mathbf{z}) - \Psi_\gamma(\mathbf{z}'), \mathbf{z} - \mathbf{z}' \rangle \geq \beta_\gamma \|\Psi_\gamma(\mathbf{z}) - \Psi_\gamma(\mathbf{z}')\|^2. \tag{$\natural$}$$

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It is enough to show that Ψ_γ is β_γ-cocoercive! That is,

$$\langle \Psi_\gamma(\mathbf{z}) - \Psi_\gamma(\mathbf{z}'), \mathbf{z} - \mathbf{z}' \rangle \geq \beta_\gamma \|\Psi_\gamma(\mathbf{z}) - \Psi_\gamma(\mathbf{z}')\|^2. \tag{$\natural$}$$

- Recall that $\Psi_\gamma(\mathbf{z}) := \mathbf{M}^* \Phi_\gamma(\mathbf{M}\mathbf{z})$.
- RHS of ($\natural$): We have

$$\beta_\gamma \|\mathbf{M}^* \Delta\mathbf{x}\|^2 = \beta_\gamma \langle \mathbf{M}\mathbf{M}^* \Delta\mathbf{x}, \Delta\mathbf{x} \rangle,$$

where

$$\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}' = \Phi_\gamma(\mathbf{M}\mathbf{z}) - \Phi_\gamma(\mathbf{M}\mathbf{z}').$$

- Lower bound for LHS of ($\natural$): Using semimonotonicity of $\mathbf{A}$,

$$\langle \Psi_\gamma(\mathbf{z}) - \Psi_\gamma(\mathbf{z}'), \mathbf{z} - \mathbf{z}' \rangle = \langle \mathbf{M}^* \Delta \mathbf{x}, \Delta \mathbf{z} \rangle \geq \langle \widehat{\mathbf{\Xi}}_\gamma \Delta \mathbf{x}, \Delta \mathbf{x} \rangle,$$

- Meanwhile, by the Lemma,

$$\widehat{\mathbf{\Xi}}_\gamma - \beta_\gamma \mathbf{M}\mathbf{M}^* \succeq 0.$$

Convergence of the Shadow Sequence

Theorem (Part 2)

The following hold:

5. If $\{\mathbf{x}^k\}$ is bounded and $\mathbf{U} = 0$, then there exists a solution $\mathbf{x}^* \in \operatorname{zer}(\mathbf{A} + N_{\boldsymbol{\Omega}})$ such that $\{\mathbf{x}^k\}$ converges weakly to $\mathbf{x}^*$.
6. If $\mathbf{U}$ is strongly monotone on $\boldsymbol{\Omega}$, then $\{\mathbf{x}^k\}$ converges strongly to $\mathbf{x}^*$.

Specialization to multioperator inclusion

Suppose

- $\mathbf{A} = \operatorname{diag}(A_1, \ldots, A_n)$ where A_i is maximal $(\mu_i \operatorname{Id}, \nu_i \operatorname{Id})$-semimonotone.
- $\mathbf{L} = [L_{ij}]$ with $L_{ii} = \delta_i \operatorname{Id}$, $\delta_i > 0$.
- $\mathbf{M} = M \otimes \operatorname{Id}$ with *any* $M \in \mathbb{R}^{n \times m}$ such that $\ker(M^\top) = \operatorname{span}\{(1, \ldots, 1)\}$.
- Denote

$$\sigma := \frac{2\delta_r(-\nu_r)_+}{1 + \sqrt{1 - 4\mu_r\nu_r}} \quad \& \quad \varsigma := \min\left\{ \frac{-\delta_i(1 + \sqrt{1 - 4\mu_i\nu_i})}{2\mu_i} : \mu_i < 0 \right\}$$

where $\nu_r = \min_i \nu_r$.

<table>
<tr><th rowspan="2">$\Lambda \cap \Gamma$</th><th rowspan="2">$\mu_i = 0 \quad (\forall i)$</th><th colspan="2">$\sum_{i=1}^n \mu_i > 0$</th></tr>
<tr><th>$\mu_i \geq 0$</th><th>$\exists i$ s.t. $\mu_i < 0$</th></tr>
<tr><td>$\nu_i = 0 \quad (\forall i)$</td><td colspan="2" rowspan="2">$(0, +\infty)$</td><td rowspan="2">$(0, \varsigma)$</td></tr>
<tr><td>$\nu_i \geq 0 \quad (\forall i)$</td></tr>
<tr><td>$\exists j$ s.t. $\nu_j < 0$
$\nu_i > 0 \quad (\forall i \neq j)$
$\sum_{i=1}^n \nu_i^{-1} < 0$</td><td colspan="2">$(\sigma, +\infty)$</td><td>$(\sigma, \varsigma)$</td></tr>
</table>

Comparison with existing work on **monotone setting**

1. Dao, Tam, Truong (2026) proposed the algorithm

$$\begin{cases} x_i^k \in J_{\gamma D_i^{-1} A_i}\left(D_i^{-1} \sum_{j=1}^m M_{ij} z_j^k + D_i^{-1} \sum_{j=1}^n N_{ij} x_j^k - \gamma D_i^{-1} \sum_{j=1}^p P_{ij} B_j \left(\sum_{l=1}^n R_{jl} x_l^k\right)\right), \\ \qquad\qquad (\text{for } i = 1, 2, \dots, n), \\ z_i^{k+1} = z_i^k - \lambda_k \sum_{j=1}^n M_{ji} x_j^k, \qquad (\text{for } i = 1, 2, \dots, m). \end{cases}$$

   for solving $\sum_i A_i + \sum_j B_j$
   (So far, the most general algorithm for this setting.)

2. This algorithm can be recovered by our framework.

3 Comparisons

- **Coefficients:** M_{ij}, N_{ij}, D_i, P_{ij} and R_{ij} of DTT must be *scalars*.
- **Condition for convergence:** They require $\mathbf{L}_s \succeq \frac{1}{2}\mathbf{M}\mathbf{M}^*$ which implies our conditions:
  - $\ker(\mathbf{L}_s) = \mathbf{\Omega}$.
  - $\mathbf{L}_s$ is strongly monotone on $\mathbf{\Omega}^\perp$.
- **Stepsize γ and parameter λ_k.** Ours allow wider range of γ and λ_k.
- **Proof technique.** Ours do not rely on the specific componentwise structure of the algorithm.

Comparisons: **Nonmonotone setting**

1. **Multioperator setting:** Can only compare with our two papers on the *specific* product space DR.
   - Generalized monotone. We do not require $\mu_n \neq 0$.
   - Comonotone. We don't require that $\mu_n < 0$; the negative modulus can be any of the μ_i's.
   - Our analysis is unified via a general framework.

Comparisons: **Nonmonotone setting**

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   - Comonotone. We don't require that $\mu_n < 0$; the negative modulus can be any of the μ_i's.
   - Our analysis is unified via a general framework.
2. **Two-operator setting:**
   - We recover the result of Evens–Pas–Latafat–Patrinos (2025).
   - We extend the convergence result to infinite-dimensional case.

Concluding remarks

Main message

Multioperator splitting algorithms can be analyzed in one general framework, and their convergence can be extended to *semimonotone settings*.

Concluding remarks

Main message

Multioperator splitting algorithms can be analyzed in one general framework, and their convergence can be extended to *semimonotone settings*.

- Our present work considers a more general model

$$0 \in \mathbf{A}\mathbf{x} + \mathbf{P}\mathbf{B}(\mathbf{R}\mathbf{x}) + N_{\boldsymbol{\Omega}_{\mathbf{b}}}(\mathbf{x})$$

which includes forward terms $\mathbf{B}$. Here,

$$\boldsymbol{\Omega}_{\mathbf{b}} = \{\mathbf{x} \in \boldsymbol{\mathcal{H}} : \mathbf{M}^*\mathbf{x} = \mathbf{b}\}.$$

Motivating problems as inclusions

Optimization model	Inclusion form
$\min_x \sum_{i=1}^m f_i(x) + \sum_{j=1}^p h_j(x)$	$0 \in \sum_{i=1}^m A_i(x) + \sum_{j=1}^p B_j(x)$
$\min\limits_x \sum\limits_{i=1}^m f_i(x) + \sum\limits_{k=1}^r g_k(S_k x) + \sum\limits_{j=1}^p h_j(x)$	$0 \in \sum\limits_{i=1}^m A_i(x) + \sum\limits_{k=1}^r S_k^* C_k(S_k x) + \sum\limits_{j=1}^p B_j(x)$
$\min\limits_{x_1,\ldots,x_m} \sum\limits_{i=1}^m f_i(x_i)$ s.t. $\sum\limits_{i=1}^m S_i x_i = b$	$0 \in A_1(x_1) \times \cdots \times A_n(x_n) + N_{\Omega_b}(x)$, $\Omega_b = \left\{ x : \sum\limits_{i=1}^m S_i x_i = b \right\}$

Thank you for listening!

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Appendix: Main idea to prove 5

- From part 3: $\mathbf{z}^k \rightharpoonup \mathbf{z}^*$.
- From part 2: $\mathbf{M}^*\mathbf{x}^k \to 0$. This implies

$$\Pi_{\boldsymbol{\Omega}^\perp}\mathbf{x}^k \to 0$$

for any $\bar{\mathbf{x}} \in \boldsymbol{\Omega}$.

- Rewrite

$$\mathbf{M}\mathbf{z}^k + \in \gamma\mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k \qquad \Longleftrightarrow \qquad \mathbf{x}^k \in \Phi_\gamma(\mathbf{M}\mathbf{z}^k)$$

equivalently as

$$\mathbf{x}^k \in \mathbf{A}^{-1}\left(\frac{1}{\gamma}\underbrace{\left(\mathbf{M}\mathbf{z}^k - \Pi_{\boldsymbol{\Omega}^\perp}\mathbf{L}\mathbf{x}^k\right)}_{\in\boldsymbol{\Omega}^\perp} - \frac{1}{\gamma}\Pi_{\boldsymbol{\Omega}}\mathbf{L}\Pi_{\boldsymbol{\Omega}^\perp}\mathbf{x}^k\right)$$

Appendix: Main idea to prove 5 (2)

Lemma (see Bauschke & Combettes, 2017)

Let $\boldsymbol{\Omega} \subseteq \boldsymbol{\mathcal{H}}$ be a closed linear subspace, $\bar{\mathbf{x}} \in \mathcal{H}$ and $\mathbf{T} : \boldsymbol{\mathcal{H}} \rightrightarrows \boldsymbol{\mathcal{H}}$ be maximally monotone. Let $\mathbf{v}^k \in \mathbf{T}\mathbf{u}^k$ and suppose that

$$\mathbf{u}^k \rightharpoonup \mathbf{u}^*, \quad \mathbf{v}^k \rightharpoonup \mathbf{v}^*, \quad \Pi_{\boldsymbol{\Omega}}\mathbf{u}^k \to 0, \quad \text{and} \quad \Pi_{\boldsymbol{\Omega}^\perp}(\mathbf{v}^k - \bar{\mathbf{x}}) \to 0, \tag{1}$$

then $\mathbf{v}^* \in \mathbf{T}\mathbf{u}^*$.

- If $\mathbf{x}^{k_j} \rightharpoonup \mathbf{x}^*$, then $\mathbf{x}^* \in \Phi_\gamma(\mathbf{M}\mathbf{z}^*)$.
- Since Φ_γ is single-valued,

$$\text{all accumulation points of } \{\mathbf{x}^k\} \text{ are equal to } \Phi_\gamma(\mathbf{M}\mathbf{z}^*).$$

Example 1: $A_1 + A_2 + B_1$

We want to solve

$$0 \in A_1 u + A_2 u + B_1 u.$$

Lift the variable:

$$\mathbf{x} = (x_1, x_2) \in \mathcal{H}^2, \qquad \mathbf{A}\mathbf{x} = \begin{bmatrix} A_1 x_1 \\ A_2 x_2 \end{bmatrix},$$

To recover the original problem, impose consensus:

$$\boldsymbol{\Omega}_{\mathbf{b}} := \{(x_1, x_2) \in \mathcal{H}^2 : \; x_1 = x_2\},$$

which is a subspace, so $\mathbf{b} = 0$. We may choose $\mathbf{M} : \mathcal{H} \to \mathcal{H}^2$, as

$$\mathbf{M} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} \\ -\mathrm{Id}_{\mathcal{H}} \end{bmatrix}.$$

Example 1: Choosing $\mathbf{P}, \mathbf{R}, \mathbf{L}$

Choose

$$\mathbf{R} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} & 0 \end{bmatrix}, \qquad \mathbf{P} = \begin{bmatrix} 0 \\ \mathrm{Id}_{\mathcal{H}} \end{bmatrix}.$$

Thus

$$\mathbf{P}\mathbf{B}(\mathbf{R}\mathbf{x}) = \begin{bmatrix} 0 \\ B_1 x_1 \end{bmatrix}.$$

Finally, take

$$\mathbf{L} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} & 0 \\ -2\,\mathrm{Id}_{\mathcal{H}} & \mathrm{Id}_{\mathcal{H}} \end{bmatrix}, \qquad \mathbf{q} = 0.$$

The proposed algorithm uses

$$\mathbf{M}\mathbf{z}^k \in \gamma\mathbf{A}\mathbf{x}^k + \gamma\mathbf{P}\mathbf{B}(\mathbf{R}\mathbf{x}^k) + \mathbf{L}\mathbf{x}^k.$$

Example 1: Computing the iteration

Substituting the choices gives

$$\begin{bmatrix} z^k \\ -z^k \end{bmatrix} \in \gamma \begin{bmatrix} A_1 x_1^k \\ A_2 x_2^k \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ B_1 x_1^k \end{bmatrix} + \begin{bmatrix} x_1^k \\ -2x_1^k + x_2^k \end{bmatrix}.$$

Equivalently,

$$\begin{cases} z^k \in \gamma A_1 x_1^k + x_1^k, \\ 2x_1^k - z^k - \gamma B_1 x_1^k \in \gamma A_2 x_2^k + x_2^k. \end{cases}$$

Hence

$$\begin{cases} z^k \in (\gamma A_1 + \mathrm{Id}) x_1^k, \\ 2x_1^k - z^k - \gamma B_1 x_1^k \in (\gamma A_2 + \mathrm{Id}) x_2^k. \end{cases}$$

Example 1: Resulting algorithm

Thus,

$$\begin{cases} x_1^k \in J_{\gamma A_1}(z^k), \\ x_2^k \in J_{\gamma A_2}\big(2x_1^k - z^k - \gamma B_1 x_1^k\big), \\ z^{k+1} = z^k - \lambda_k(x_1^k - x_2^k). \end{cases}$$

That is,

$$z^{k+1} = z^k + \lambda_k(x_2^k - x_1^k).$$

Recovered method

This realization gives the Davis–Yin splitting for

$$0 \in A_1 u + A_2 u + B_1 u.$$

Example 2: $A_1 + A_2 + S_1^* C_1 S_1 + B_1$

We want to solve

$$0 \in S_1^* C_1(S_1 u) + B_1 u.$$

Lift the variable:

$$\mathbf{x} = (x_1, x_2, v) \in \mathcal{H} \times \mathcal{H} \times \mathcal{K}, \qquad \mathbf{A}\mathbf{x} = \begin{bmatrix} A_1 x_1 \\ A_2 x_2 \\ C_1 v \end{bmatrix}.$$

To recover the original problem, impose

$$\boldsymbol{\Omega}_{\mathbf{b}} := \{(x_1, x_2, v) : x_1 = x_2,\ v = S_1 x_1\},$$

which is a subspace, so $\mathbf{b} = 0$. We may choose $\mathbf{M} : \mathcal{H} \times \mathcal{K} \to \mathcal{H} \times \mathcal{H} \times \mathcal{K}$ as

$$\mathbf{M} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} & -S_1^* \\ -\mathrm{Id}_{\mathcal{H}} & 0 \\ 0 & \mathrm{Id}_{\mathcal{K}} \end{bmatrix} \qquad \text{with } \boldsymbol{\Omega}^{\perp} = \operatorname{ran}(\mathbf{M}) = \{(a, b, c) : a + b + S_1^* c = 0\}.$$

Then

$$\mathbf{M}^* \mathbf{x} = \begin{bmatrix} x_1 - x_2 \\ v - S_1 x_1 \end{bmatrix}.$$

Example 2: Choosing **P**, **R**, **L**

Choose

$$\mathbf{R} = \begin{bmatrix} \mathrm{Id}_{\mathcal{H}} & 0 & 0 \end{bmatrix}, \qquad \mathbf{P} = \begin{bmatrix} 0 \\ \mathrm{Id}_{\mathcal{H}} \\ 0 \end{bmatrix}.$$

Thus

$$\mathbf{P}\mathbf{B}(\mathbf{R}\mathbf{x}) = \begin{bmatrix} 0 \\ B_1 x_1 \\ 0 \end{bmatrix}.$$

Finally, take

$$\mathbf{L} = \begin{bmatrix} \alpha\,\mathrm{Id}_{\mathcal{H}} & 0 & 0 \\ 2\beta S_1^* S_1 - 2\alpha\,\mathrm{Id}_{\mathcal{H}} & \alpha\,\mathrm{Id}_{\mathcal{H}} & 0 \\ -2\beta S_1 & -2\beta S_1 & 2\beta\,\mathrm{Id}_{\mathcal{K}} \end{bmatrix}, \qquad \mathbf{q} = 0,$$

where

$$\beta > 0, \qquad \alpha > \frac{\beta \|S_1\|^2}{2}. \quad \text{(Important for convergence)}$$

Example 2: Computing the iteration

The proposed algorithm uses

$$\mathbf{M}\mathbf{z}^k \in \gamma \mathbf{A}\mathbf{x}^k + \gamma \mathbf{P}\mathbf{B}(\mathbf{R}\mathbf{x}^k) + \mathbf{L}\mathbf{x}^k, \qquad \mathbf{z}^k = (z^k, w^k).$$

Substituting the choices gives

$$\begin{bmatrix} z^k - S_1^* w^k \\ -z^k \\ w^k \end{bmatrix} \in \gamma \begin{bmatrix} A_1 \\ A_2 \\ C_1 v^k \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ B_1 x_1^k \\ 0 \end{bmatrix} + \mathbf{L} \begin{bmatrix} x_1^k \\ x_2^k \\ v^k \end{bmatrix}.$$

Equivalently,

$$\begin{cases} \dfrac{1}{\alpha}\left(z^k - S_1^* w^k\right) = \dfrac{\gamma}{\alpha} A_1 x_1^k + x_1^k, \\ 2x_1^k - \dfrac{1}{\alpha}\left(z^k + \gamma B_1 x_1^k + 2\beta S_1^* S_1 x_1^k\right) = \dfrac{\gamma}{\alpha} A_2 x_2^k + x_2^k, \\ S_1 x_1^k + S_1 x_2^k + \dfrac{1}{2\beta} w^k \in \left(\dfrac{\gamma}{2\beta} C_1 + \mathrm{Id}\right) v^k. \end{cases}$$

Example 2: Resulting iteration

Thus,

$$
\begin{cases}
x_1^k = J_{\frac{\gamma}{\alpha}A_1}\left(\frac{1}{\alpha}\left(z^k - S_1^* w^k\right),\right) \\
x_2^k = J_{\frac{\gamma}{\alpha}A_2}\left(2x_1^k - \frac{1}{\alpha}\left(z^k + \gamma B_1 x_1^k + 2\beta S_1^* S_1 x_1^k\right)\right), \\
v^k = J_{\frac{\gamma}{2\beta}C_1}\left(S_1 x_1^k + S_1 x_2^k + \frac{1}{2\beta} w^k\right), \\
z^{k+1} = z^k - \lambda_k (x_1^k - x_2^k), \\
w^{k+1} = w^k - \lambda_k (v^k - S_1 x_1^k).
\end{cases}
$$

Example 3: two-block affine constraint

Consider

$$0 \in A_1 x_1 \times A_2 x_2 + N_{\mathbf{\Omega}_b}(x_1, x_2), \qquad \mathbf{\Omega}_b = \{(x_1, x_2) : S_1 x_1 + S_2 x_2 = b\}.$$

Take

$$\mathbf{M} z = \begin{bmatrix} S_1^* z \\ S_2^* z \end{bmatrix}, \qquad \mathbf{M}^* \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = S_1 x_1 + S_2 x_2,$$

and choose

$$\mathbf{L} = \begin{bmatrix} S_1^* S_1 & 0 \\ 2 S_2^* S_1 & S_2^* S_2 \end{bmatrix}, \qquad \mathbf{q} = \begin{bmatrix} 0 \\ S_2^* b \end{bmatrix}.$$

For $\mathbf{\Omega} = \ker \mathbf{M}^*$, this choice satisfies

$$\mathbf{L}(\mathbf{\Omega}) \subseteq \mathbf{\Omega}^{\perp},$$

since if $S_1 x_1 + S_2 x_2 = 0$, then

$$\mathbf{L} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} S_1^* S_1 x_1 \\ S_2^* S_1 x_1 \end{bmatrix} \in \operatorname{ran} \mathbf{M}.$$

Example 3: Computing the iteration

The first step is

$$\mathbf{M}z^k + \mathbf{q} \in \gamma \mathbf{A}\mathbf{x}^k + \mathbf{L}\mathbf{x}^k.$$

Equivalently,

$$\begin{bmatrix} S_1^* z^k \\ S_2^* z^k + S_2^* b \end{bmatrix} \in \gamma \begin{bmatrix} A_1 x_1^k \\ A_2 x_2^k \end{bmatrix} + \begin{bmatrix} S_1^* S_1 x_1^k \\ 2 S_2^* S_1 x_1^k + S_2^* S_2 x_2^k \end{bmatrix}.$$

Thus,

$$\begin{cases} S_1^* z^k \in \gamma A_1 x_1^k + S_1^* S_1 x_1^k, \\ S_2^*(z^k + b) \in \gamma A_2 x_2^k + 2 S_2^* S_1 x_1^k + S_2^* S_2 x_2^k, \\ z^{k+1} = z^k - \lambda_k (S_1 x_1^k + S_2 x_2^k - b). \end{cases}$$

Example 3: Special case – two-block optimization

For

$$\min_{x_1, x_2} f_1(x_1) + f_2(x_2) \qquad \text{s.t.} \qquad S_1 x_1 + S_2 x_2 = b,$$

take $A_i = \partial f_i$. Then the iteration becomes

$$\begin{cases} x_1^k \in \arg\min_{x_1} \left\{ f_1(x_1) + \dfrac{1}{2\gamma} \left\| S_1 x_1 - z^k \right\|^2 \right\}, \\ x_2^k \in \arg\min_{x_2} \left\{ f_2(x_2) + \dfrac{1}{2\gamma} \left\| S_2 x_2 + 2 S_1 x_1^k - b - z^k \right\|^2 \right\}, \\ z^{k+1} = z^k - \lambda_k (S_1 x_1^k + S_2 x_2^k - b). \end{cases}$$